

Coverpoint / bins

Functional coverage asks whether you exercised interesting values, not just whether the test passed

Coverpoints, bins, holes

- A covergroup wraps one or more coverpoints
- Each coverpoint names a signal or slice, here data three colon zero, the low four bits
- Bins group values: low covers zero through three, mid covers four through seven, and so on
- When you sample a value, every bin that contains it gets a hit
- A hole is a defined bin with zero hits after your tests
- This lab stays with single coverpoints

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Coverpoint / bins sketch

Document-aid literacy: a **coverpoint** partitions values into **bins**; sampling hits bins and leaves **holes**. Not a simulator coverage DB. Starter: nibble coverpoint with `mid` already hit.

Starter example: nibble coverpoint with bins `low/mid/high/top`; sample 5 already hit `mid`. Sample more values to close holes.

Load starter example

Challenges 0 / 22 cleared

Quiz: **coverpoint**: A coverpoint usually measures...

☐ how a field/expression's values were observed (via bins)

☐ FPGA place-and-route congestion

☐ SDF interconnect delay only

☐ the Makefile target name

Show hint

Check

Next

Idle

1

2

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Core ideas

Coverpoint

A field (or expression) you care about.

Bins

Named buckets over that field's values.

Sample

A observed value hits one bin.

Hole

A defined bin never hit yet.

Browser lab starter

learn_sv_tb — cover bins

Real SV TB track practice

- In the real track, open this module's examples prompts
- Restate coverpoint literacy in one sentence
- On paper, write a coverpoint on a two-bit opcode with idle, read, write
- List three sample sequences and mark which bins each sequence hits
- Sketch one hole you would close by adding a directed test
- Map the same bin names to a coverage section in a verification plan you have seen

Pitfalls to watch

- Do not confuse a hole with an ignorable value
- Do not assume one hundred percent bin hit means done
- Do not sample randomly without a plan
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On paper, write one coverpoint with four bins and mark which bin value seven hits
- When you are ready